Certificate proof and verification supplement

Jason Davies

November 4, 2026

This supplement preserves the computer-assisted corner proof used by the Lean development. It is independent of the elementary corner argument in the revised article and is not a formalisation of that argument.

1 Reduction to finite problems

Eight copies of any variety solve that variety, so a solution-free collection has every multiplicity at most seven. Suppose such a collection has size 15.

Lemma 1 (Anchored Hall cover)**.** *Suppose a collection contains m copies of a variety A and does not solve A. There is a set K of its target positions such that*

$$|K| \le 7 - m, \qquad \#\{\textit{other copies fitting some position outside } K\} \le 7 - m - |K|. \tag{1}$$

Conversely, such a set certifies failure of target A.

Proof. Apply (1) to A and put $K = D^c$. Every copy of A is a neighbour of the nonempty set D. Subtract these m copies from the bound $|D| - 1 = 7 - |K|$. The converse is the same Hall inequality read backwards. □

First suppose the maximum multiplicity m is at least two. Choose an adjacent-repeat anchor attaining m if possible; otherwise choose an opposite-repeat anchor attaining m. In the latter case all adjacent-repeat multiplicities are at most $m - 1$. Relabel the colours to send the anchor to a fixed representative of its orbit.

For each cover K in Lemma 1, put

$$S(K) = N_A(K^c) \setminus \{A\}, \qquad b(K) = 7 - m - |K|.$$

Write $S = S(K)$ and $b = b(K)$. The case has $x_A = m$, $x(S) \le b$, and the multiplicity bounds just described. Additional upper bounds use two-variety solutions. If j copies of A and $8 - j$ copies of V build a target, with $j \le m$, then $x_V \le 7 - j$ in a solution-free collection. These tests are finite: the two varieties must cover all eight target positions, A must fit at least j, and V at least $8 - j$. They are also sufficient: at most j positions are unavailable to V, and all of them are available to A. Assign these to A, extend the assignment to j positions that A fits, and assign the remaining $8 - j$ positions to V.

Colour permutations and simultaneous reflection of the entire collection preserve solvability. First identify equal pairs $(S(K), b(K))$, even when they arise from inequivalent position subsets K; then retain one representative per orbit of these pairs under the anchor stabiliser. The counts below refer to these constraint pairs. Transporting a cover also transports its constrained varieties, budget and multiplicities, so no counterexample is lost. Global reflection transports the whole collection; individual cubes still move only by proper rotations. The repeated-variety cases give 244 constraint orbits; the breakdown is in Appendix 3.

For distinct collections a different choice of anchor reduces the count.

Lemma 2 (Shared two-colour corner)**.** *Among eleven adjacent-repeat cubes, two share a two-colour corner type. If a distinct collection contains such a pair and one member is chosen as anchor, the set consisting of its two two-colour positions is not Hall-deficient.*

Proof. Each adjacent-repeat cube with repeated colour r has exactly two such types, rrx and rry, with $x \neq y$. There are only $5 \cdot 4 = 20$ types rrx. Eleven cubes contribute 22 incidences, so two cubes share a type. For the two characteristic positions of the chosen anchor, the anchor itself and the other cube are two distinct neighbours. Their neighbourhood therefore has size at least two. □

If an opposite-repeat variety is present, choose it as anchor; this gives 31 constraint orbits with every multiplicity at most one. Otherwise use Lemma 2 to choose an adjacent-repeat anchor. Omit the cover complementary to its two characteristic positions and set all opposite-repeat capacities to zero. Its stabiliser preserves these positions, so the omission is compatible with the orbit reduction. It removes one of the 83 adjacent-repeat constraint orbits, leaving 82 further cases.

Proposition 3. *Every solution-free fifteen-cube collection is carried by colour relabelling and, if needed, global reflection into one of the 357 exported root cases.*

Proof. The preceding arguments cover repeated and distinct collections. The finite catalogue, two-variety bounds and coverage of all anchor-cover orbits are checked from physical rotations by the accompanying audit. It regenerates rotations from quarter turns, separately from the signed-axis permutations used by the case generator, and checks every exported bound and every cover orbit. The resulting counts are

$$244 + 31 + 82 = 357.$$

The cases need not be disjoint; completeness of the cover is what is required. □

Each root specifies an anchor A of multiplicity m, coordinate bounds ℓ, u, a set S of non-anchor varieties and a budget b. The bounds fix $\ell_A = u_A = m$, with $\ell_i = 0$ for $i \neq A$. A vector $x \in \mathbb{Z}^{75}_{\geq 0}$ is *admissible* for this root if

$$\ell \leq x \leq u, \qquad \sum_{i=1}^{75} x_i = 15, \qquad x(S) \leq b, \tag{2}$$
$$\text{for every target } T, \text{ there is } \varnothing \neq D \subseteq \{1, \ldots, 8\} \text{ with } x(N_T(D)) \leq |D| - 1.$$

Thus admissibility includes failure of every target, in addition to the anchor constraints.

The cover can be reduced without further search. If $S' \subseteq S$ and $b' \geq b$, then $x(S) \leq b$ implies $x(S') \leq b'$ for nonnegative x. Together with coordinatewise inclusion of the root boxes, this identifies 49 cases contained in other cases, leaving 308. Write u for the root upper bounds. In 29 retained cases, $b + u(S^c) < 15$ already contradicts the prescribed total, so only 279 cases need further exclusion proofs. The exact map and retained case numbers are in `data/q5-root-reduction.json`; Lean checks all containments and arithmetic exclusions. The archive retains all 357 original cases.

Lemma 4 (Hall-budget projection)**.** *Suppose a root has anchor multiplicity m, budget $x(S) \leq b$, and total 15, with the anchor outside S. For $0 \leq \lambda \leq b$, put $G = 15 - b + \lambda$ and suppose $G \geq m$. Every admissible vector x has a vector $y \leq x$ with the same anchor multiplicity, total G, budget $y(S) \leq \lambda$, and a Hall deficiency for every target. It obeys the original upper bounds, tightened to $\min(u_i, \lambda)$ on S.*

Proof. First retain only $\min(x(S), \lambda)$ copies in S. This removes at most $b - \lambda$ copies and leaves the anchor untouched, so the remaining total is at least G. Delete further non-anchor copies until the total is G; this is possible because $G \geq m$. Every coordinate decreases, so all upper bounds and Hall deficiency inequalities are preserved. Nonnegativity and $y(S) \leq \lambda$ give $y_i \leq \lambda$ for $i \in S$. □

For our anchored covers, $b = 7 - m - |K|$, so

$$G = m + 8 + |K| + \lambda.$$

Thus the reduced collection consists of the m fixed anchor copies and $8 + |K| + \lambda$ other copies; in particular, $G \geq m$ is automatic. Apply the lemma after selecting and normalising the original root, retaining its bounds. No new anchor selection is required for the smaller collection. The resulting 279 projected problems have totals between 9 and 15. Their parameters and exceptional cases are recorded in Appendix 3.

2 Interval certificates

For a projected problem, admissibility is defined by (2) with tightened upper bounds and total G. In this section denote its projected budget by b. We certify that no admissible vector exists.

Write a box as $\ell \leq x \leq u$. For a cut $x(S) \leq r$, two necessary conditions for feasibility in the box are

$$\ell(S) \leq r, \qquad G \leq r + u(S^c). \tag{3}$$

The first follows from the lower bounds; the second follows from $G = x(S) + x(S^c)$. When they hold, put $\sigma = r - \ell(S)$ and $\gamma = r + u(S^c) - G$. Every satisfying vector obeys

$$x_i \leq \min(u_i, \ell_i + \sigma) \quad (i \in S), \qquad x_i \geq \max(\ell_i, u_i - \gamma) \quad (i \notin S). \tag{4}$$

For example, the upper bound follows by giving all other coordinates of S their lower bounds; the lower bound follows by giving all other coordinates of S^c their upper bounds.

For a target, retain its cuts satisfying (3). If none remain, no vector in the box can fail that target. Otherwise let I and U be the intersection and union of their neighbourhoods, and let σ_*, γ_* be the largest corresponding σ, γ. Then all vectors in the box that fail this target remain in the refined box

$$u_i' = \begin{cases} \min(u_i, \ell_i + \sigma_*) & i \in I, \\ u_i & i \notin I, \end{cases} \qquad \ell_i' = \begin{cases} \max(\ell_i, u_i - \gamma_*) & i \notin U, \\ \ell_i & i \in U. \end{cases} \tag{5}$$

Indeed, a failing vector satisfies one of the retained cuts, and its bounds in (4) imply (5). The root budget is treated as a family containing a single cut. The total $\sum_i x_i = G$ gives

$$\ell_i' = \max\left(\ell_i, G - \sum_{j \neq i} u_j\right), \qquad u_i' = \min\left(u_i, G - \sum_{j \neq i} \ell_j\right).$$

A total-bound leaf closes exactly when $\sum_i \ell_i > G$ or $\sum_i u_i < G$. A budget leaf closes when $\ell(S) > b$ or $b + u(S^c) < G$; a target leaf closes when none of its 255 cuts satisfies (3). The checker thus has 77 closing reasons: one total test, one budget test and 75 target tests. Refinements must tighten at least one bound and leave nonempty coordinate intervals; a split must satisfy $\ell_i \leq v < u_i$.

A certificate is a finite tree of refinements and integer splits $x_i \leq v$ or $x_i \geq v + 1$. Each leaf is excluded by total bounds or by a cut family with no surviving cut. Induction on this tree proves that its root contains no admissible vector. The separate C++ checker reads these instructions and verifies the deductions; it performs no search and does not use the search program's matching implementation.

Proposition 5 (Certificate replay)**.** *The separate checker accepts an exclusion certificate for each of the 279 projected problems.*

Archive sizes and rejection controls are recorded in Appendix 3.

Alternative proof of Theorem 3. Lemma 12 gives $N_5 \geq 15$. A solution-free collection of fifteen cubes would satisfy an exported root case by Proposition 3, and would supply a Hall deficiency for every target. Containment and the initial budget tests leave one of 279 roots. Lemma 4 supplies an admissible vector in its projected problem, which Proposition 5 and the sound interval deductions exclude. Hence $N_5 \leq 15$. □

3 Verification and reproduction details

3.1 Finite certificates

The repeated-variety cases have the following counts, totalling 244.

Anchor / maximum multiplicity	2	3	4	5	6	7
Adjacent-repeat	72	55	31	14	4	1
Opposite-repeat	26	21	11	6	2	1
Total	98	76	42	20	6	2

Of the 279 remaining roots, 277 admit an exclusion certificate with $\lambda = 0$. The other two, original cases 68 and 69, use $\lambda = 1$ and $G = 12$. Their $\lambda = 0$, $G = 11$ problems actually have solution-free collections, supplied and checked by physical matching in the projected manifest. The projected totals range from 9 to 15. All 101 roots whose original budget was zero retain their original certificates.

The projected archive contains 499,829 instructions, 95.69% fewer than the original archive. Both archives and the search computations are documented in the repository. Every archived header is checked against its exported root case. Eighty negative controls are rejected, including all 77 possible direct closing claims against a separate $G = 11$ fixture, not a $G = 15$ root problem. This fixture is a known solution-free eleven-cube collection: it satisfies its total, fixed anchor count, coordinate bounds and zero root budget, and fails all 75 targets by physical matching. It is `data/q5-distinct-known-control.txt`; its regeneration and witness checks are in `scripts/check_five_colour_distinct.py`.

3.2 Lean implementation

The development uses Lean 4 and mathlib [7, 8], both at `v4.35.0`. It proves the physical theorems, including the connection between cube arrangements and the finite certificate problems. The physical model and palette-independent geometry come from the shared `colour-cubes-core` library [6].

Lean proves completeness and uniqueness of the 75 varieties and checks all 45,000 corner-compatibility bits using the shared corner–rotation theorem. The lower bound uses 75 deficient corner sets: 72 singletons and three sets of sizes 5, 6 and 6. Its theorem is `FiveColour.Physical.lower_bound` in `FiveColour/LowerBound.lean`. In the formal proof, this replaces the elementary target classification in Lemma 12; catalogue completeness and rotation transport extend the result to every physical target.

The upper-bound formalisation follows the Hall reduction, symmetry transport, anchor selection and shared-corner argument. Finite checks verify capacities, cover representatives and the reduction from 357 roots to 279. The deletion step is `FiveColour.Physical.root_projection` in `FiveColour/RootProjection.lean`. The bridge to the archive, in `FiveColour/`

`ProjectedBridge.lean`, identifies the full admissibility predicate: coordinate bounds, total, budget and all Hall-cut families.

For efficient replay, Lean proves soundness of a packed implementation of the interval checker. For multiplicities $0 \le x_i \le 7$, index the varieties by $0, \dots, 74$ and put

$$P(x) = \sum_{i=0}^{74} x_i B^i, \qquad W(S) = \sum_{i \in S} B^{74-i}, \qquad B = 1024.$$

Then

$$x(S) = \left\lfloor \frac{P(x)W(S)}{B^{74}} \right\rfloor \bmod B.$$

Every coefficient in the product is at most $75 \cdot 7 < 1024$, so no carry can affect this coefficient extraction. Lean proves the identity, decoding, coordinate updates and equivalence with the integer inequalities, and checks all 19,125 neighbourhood weights.

The 279 certificates are split into 4,786 chunks of at most 128 instructions. Composition requires exact boundary-vector agreement and a proof for every referenced box; the exporter is untrusted. All chunks and their final assembly have passed kernel checking. The case mapping and certificate hashes are in `data/q5-projected-root-manifest.json`. The generated module `FiveColour/ProjectedChunks/Threshold.lean` proves the unconditional theorems `threshold15` and `sharpThreshold15`. Both lie in namespace `FiveColour.Packed.ProjectedChunks`.

The cuboid proof in Lean follows Lemma 11. The cube exports, such as `frame20`, specialise the cuboid theorems. The equivalences in `FiveColour/CubeBoxBridge.lean` identify the cube and equal-side cuboid position models, preserving exposed faces and physical placements. Cube completion likewise reuses rectangular completion, whose generic construction is adapted from [5]; no theorem of that paper is used. The auxiliary eleven-cube missing-pair lemma remains in `FiveColour/RingEleven.lean`, with its multiplicity argument proved without enumerating collections.

The small-support proofs check pair witnesses directly, without reducing them to five colour orbits, and use deficient corner sets for the two obstruction constructions. The modules are `SmallSupport.lean` and `MixedObstructions.lean` in the `FiveColour` directory.

The full cuboid classification is formalised in `FiveColour/BoxThresholds.lean`. In namespace `FiveColour.Physical`, `box_frame_threshold_iff` and `box_threshold_iff` give necessary and sufficient pool sizes for arbitrary side lengths at least two, assuming `FiveColour.Physical.Threshold15`. `Solution.lean` applies the completed corner theorem to discharge this assumption. In namespace `FiveColour.Puzzle`, it exports the unconditional theorems `box_frame_threshold` and `complete_box_threshold`, together with `frame20`, `frame_threshold_all`, `full_cube`, `complete_cube_threshold` and `support_threshold`.

The rectangular model and completion are in `FiveColour/BoxModel.lean`; compatibility is connected to proper rotations, and axis assignments are realised by proper rotations. `FiveColour/EightPlusThree.lean` proves the physical prescribed-family lemma. Its proof uses a failed Hall set to obtain the small cover directly, as in Lemma 11; signed projection and small matching patterns are checked in the kernel. The weighted bounds and upper deductions are in `FiveColour/WideFrames.lean`, `FiveColour/RodFrames.lean` and `FiveColour/StripFrames.lean`. The thin-box proofs formalise both concentration and switching targets, including the equality cases of the affine geometric bounds. Directional profiles are checked for the two canonical target varieties; proved colour and rotation transport covers all targets, including their corner, edge and axis labels. The small half-mask matching step uses greedy selection after single-mask checks. `FiveColour/BoxLowerBound.lean` proves the uniform three-variety obstruction for every shape and orientation. A separate C++ audit reconstructs the geometry and checks these profiles. No new interval certificates are used. The flow algorithm and its complexity bound in Section 7 are mathematical deductions, not additional Lean claims.

3.3 Reproduction

The dependency audits cover 133 declarations listed in `Audit.lean`, fourteen extension statements listed in `ExtensionsAudit.lean`, and the final replay theorems. Their dependencies are confined to `propext`, `Classical.choice` and `Quot.sound`. No `native_decide` is used. Reproduce the proof and audits with:

```
make verify-full WORKERS=2
```

The module guide is `FiveColour/README.md`.

A Mathlib-only comparator challenge states the corner problem geometrically, restating the library's definitions locally. It covers only the corner problem: the cuboid statements rest on the definitions in `FiveColour/BoxModel.lean`. Its equivalence with the development's model is proved by `FiveColour.Puzzle.hasSolution_iff` in `FiveColour/ComparatorBridge.lean`. The solution module applies the completed threshold theorem. A full external comparator pass remains unconfirmed; it is separate from the completed Lean proof.

References

[1] P. A. MacMahon, *New Mathematical Pastimes*, Cambridge University Press (1921). Digitised edition.

[2] E. Berkove, A. Eickemeyer and M. Schulman, *The 4-Color Cubes Puzzle*, Integers **18** (2018), A68. doi:10.5281/zenodo.10680216. Journal PDF.

[3] P. Hall, *On Representatives of Subsets*, Journal of the London Mathematical Society **s1-10**(1) (1935), 26–30. doi:10.1112/jlms/s1-10.37.26.

[4] J. Edmonds and R. M. Karp, *Theoretical Improvements in Algorithmic Efficiency for Network Flow Problems*, Journal of the ACM **19**(2) (1972), 248–264. doi:10.1145/321694.321699.

[5] J. Davies, *Three-colour solutions from four-colour cubes*, manuscript (2026). Accompanying repository.

[6] J. Davies, `colour-cubes-core`: palette-independent Lean definitions and proofs for face-coloured cubes, Lean library (2026). Release `v1.0.0`, `https://github.com/jasondavies/colour-cubes-core`.

[7] L. de Moura and S. Ullrich, *The Lean 4 Theorem Prover and Programming Language*, in Automated Deduction—CADE 28, LNCS **12699**, Springer (2021), 625–635. doi:10.1007/978-3-030-79876-5_37.

[8] The mathlib Community, *The Lean Mathematical Library*, in Proceedings of CPP 2020, ACM (2020), 367–381. doi:10.1145/3372885.3373824. arXiv:1910.09336.